Model-based inference of specific rewards in the lateral orbitofrontal cortex

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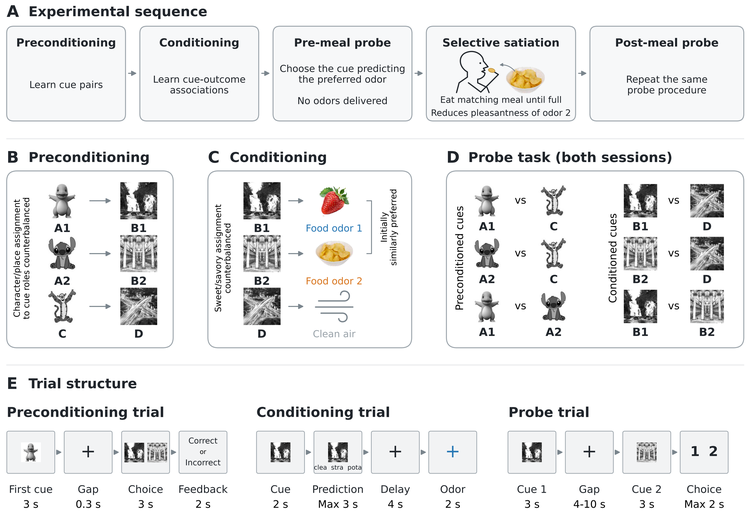
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Introduction

- **Model-based inference:** Animals can infer outcomes they have never directly experienced by combining learned associations. Sensory preconditioning (SPC) tests this ability by linking neutral cue pairs and later pairing one cue with reward [1].
- **Role of OFC:** Inactivating orbitofrontal cortex (OFC) in rats or disrupting the human OFC network selectively impairs responding to preconditioned cues, supporting its role in inference [2,3].
- **Unresolved content:** OFC represents directly learned reward identities [4], but whether preconditioned cues retrieve specific identities or scalar values remains unclear, as does how OFC represents inferred outcomes.
- **Selective devaluation:** Reducing a specific reward's value should reduce responding to its cues if they evoke that outcome's identity [5]. Responses based on a cached scalar value should be less sensitive.
- **This study:** We combined SPC, selective outcome devaluation, and fMRI to test the identity content of inferred rewards and their OFC representation, then developed a resource-limited account of the neural pattern.

Experimental design



- **Participants:** 31 adults; six visual cue pairs across two stimulus sets.
- **Schedule:** Preconditioning on Day 1; refresher training and three conditioning runs on Day 2.
- **Probe tests and ratings:** Three probe runs before and three after the meal. Odor pleasantness was rated before and after the meal.

Conclusions

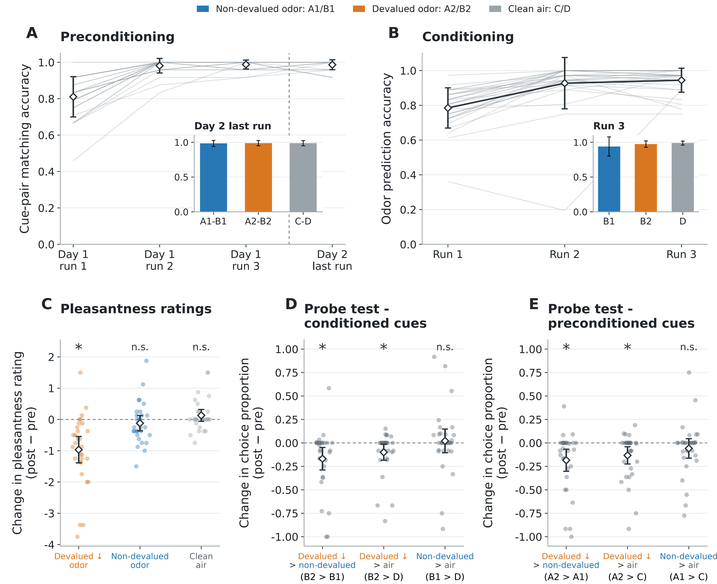
Inference retrieves specific outcome information. Devaluation changed choices for preconditioned as well as directly conditioned cues, consistent with retrieving a specific reward identity rather than only a cached value.

Lateral OFC links inferred and experienced outcomes. RSA detected inferred odor identity, and cross-decoding generalized from smelled odors to preconditioned cues, supporting a shared representational format.

Computational utility may guide representation. The resource-limited model proposes allocating more resources to outcome identity when this benefits decisions. The negative correlation between direct and inferred identity representations is consistent with a trade-off.

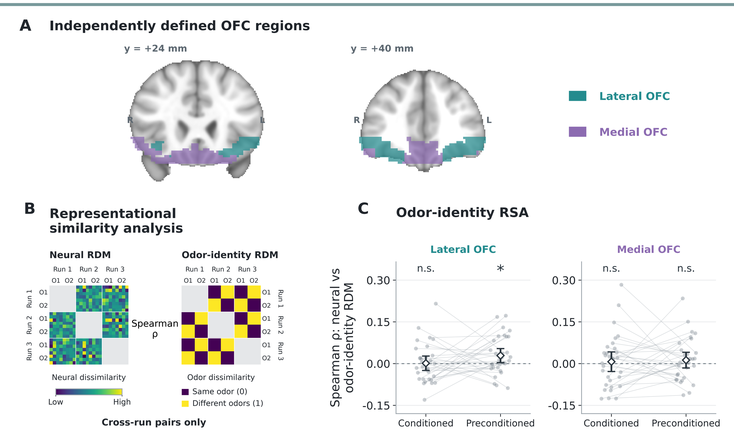
Key limits and next steps. Selectivity relative to clean-air cues was limited, and non-significant conditioned-cue effects do not establish a cue-type difference. The model is a proposed account; varying retrieval difficulty and error costs could test its predictions.

Devaluation reduces choices for conditioned and preconditioned cues



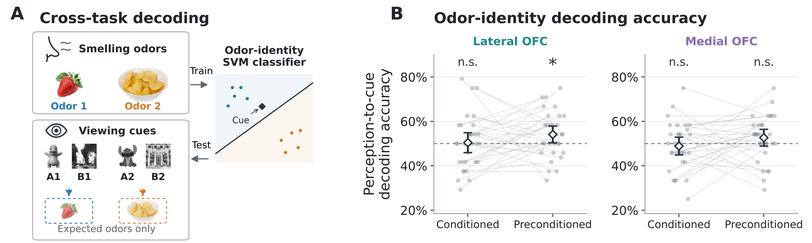
- **Analysis:** Wilcoxon signed-rank tests assessed pre-to-post changes in odor pleasantness and cue choices, and selective devaluation contrasts.
- **Finding: C:** Meal-matched odor pleasantness decreased ($p < 0.001$), more than non-matched odor ($p = 0.0013$). **D:** Conditioned-cue choices decreased for $B2 > B1$ ($p = 0.0021$) and $B2 > D$ ($p = 0.0168$). **E:** Preconditioned-cue choices decreased for $A2 > A1$ ($p = 0.0021$) and $A2 > C$ ($p = 0.0069$). These changes support specific outcome inference; selectivity relative to clean-air cues remained limited.

Lateral OFC represents inferred reward identity



- **Analysis:** Representational similarity analysis (RSA) tested odor identity in pre-meal first-cue responses from independently defined OFC regions [6], using different-run comparisons and excluding clean air.
- **Finding: C:** Lateral OFC represented inferred odor identity for preconditioned cues (mean $\rho = 0.029$, $p = 0.013$). The medial OFC effect was not significant (mean $\rho = 0.013$, $p = 0.186$). Conditioned-cue effects were not significant in either region (lateral: mean $\rho = 0.001$, $p = 0.459$; medial: mean $\rho = 0.007$, $p = 0.346$).

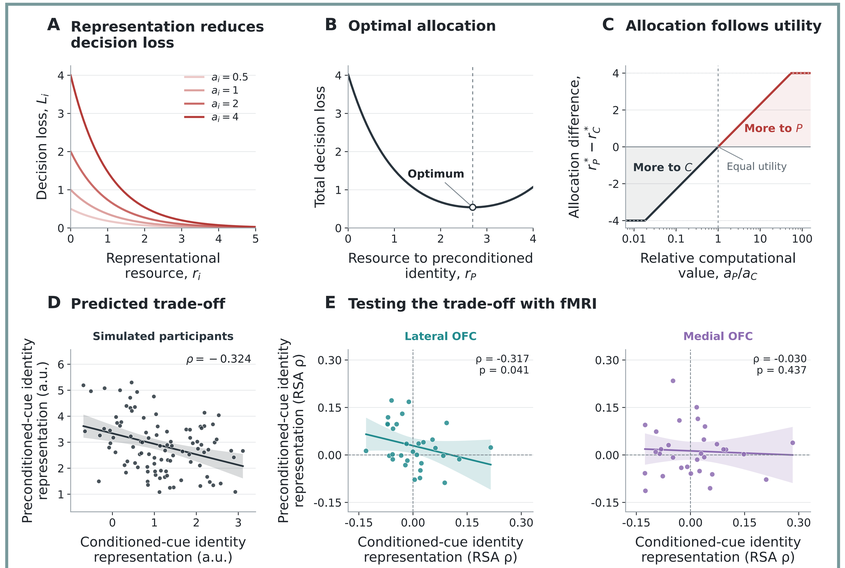
Preconditioned cues reinstate representations of experienced rewards in lateral OFC



- **Analysis:** Train on passive odor responses, then decode odor identity from pre-meal first-cue responses without odor delivery; clean-air cues excluded.
- **Finding:**

B: Preconditioned-cue decoding exceeded 50% chance in lateral OFC (54.10%, $p = 0.017$), supporting shared codes for experienced and inferred odors.
Medial OFC decoding was not significant (52.62%, $p = 0.084$).
Conditioned-cue decoding was not significant in either region (lateral: 50.40%, $p = 0.428$; medial: 48.86%, $p = 0.718$).

A resource-limited model of outcome-identity representation



- **Decision loss:** $L_i = a_i e^{-r_i}$. Increasing resource r_i reduces decision loss L_i, with diminishing returns: added resource helps less when more has already been allocated.
- **Computational value:** $a_i = q_i D_i$. Computational value is baseline retrieval-failure probability (q_i) times behavioral cost (D_i). With equal error costs, greater retrieval-failure risk for preconditioned cues implies $a_P > a_C$.
- **Shared budget:** $r_P + r_C = R$. P and C denote preconditioned and directly conditioned cues. Their nonnegative allocations share a fixed total R; more resource for one leaves less for the other.
- **Optimal allocation:** $r_P^* - r_C^* = \ln(a_P/a_C)$ for an interior optimum. The cue class with greater computational value receives more resource. Allocations are clipped to $[0, R]$.
- **Simulation (D):** Positive log-value-ratio draws enforce $a_P > a_C$ for all 100 simulated participants. With independent measurement noise, $\rho = -0.324$. Parameters are illustrative and were not fitted to fMRI data.
- **Finding (E):** Direct and inferred identity RSA scores in lateral OFC were negatively correlated ($\rho = -0.317$, $p = 0.041$).

References

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